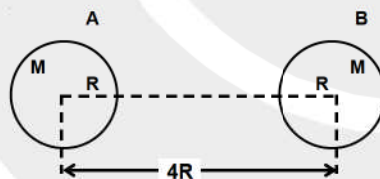


[SINGLE CORRECT CHOICE TYPE]

- Q.1 A satellite in force free space sweeps stationary interplanetary dust at a rate $(dM/dt) = \alpha v$, where M is the mass and v is the velocity of satellite and α is a constant. What is the deceleration of the satellite ?
(A) $\frac{\alpha v^2}{M}$ (B) $-\alpha v^2 / M$ (C) $-\frac{\alpha v^2}{2M}$ (D) $-\frac{2\alpha v^2}{M}$
- Q.2 The escape velocity from the earth is about 11 km/s. The escape velocity from the planet having twice the radius and the same density as the earth is
(A) 22 km/s (B) 11 km/s (C) 5.5 km/s (D) 15.5 km/s
- Q.3 A substance breaks down by a stress of 10^6 N/m^2 . If the density of the material of the wire is $3 \times 10^3 \text{ kg/m}^3$, then the length of the wire of that substance which will break under its own weight when suspended vertically is
(A) 3.4 m (B) 34 m (C) 340 m (D) none of these
- Q.4 One end of a long metallic wire of length L is tied to ceiling. The other end is tied to mass less spring of spring constant K . A mass m hangs freely from the end of spring. The area of cross-section and the Young modulus of wire are A and Y respectively. It mass is slightly pulled down and released, it will oscillate with a time period T equal to
(A) $2\pi \sqrt{m/K}$ (B) $2\pi \sqrt{m(YA + KL)/YAK}$
(C) $2\pi \sqrt{mYA/KL}$ (D) None of these
- Q.5 Two spherical massive bodies of uniform density each of mass M and radius R are kept at a distance $4R$ apart. Then the minimum speed ($V_{\min}$) required to project a particle from the surface of body A such that it will never return to the surface of the same body is

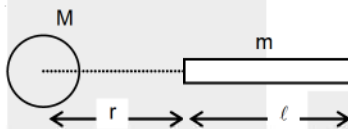


- (A) $V_{\min} = \sqrt{\frac{12GM}{5R}}$ (B) $V_{\min} = \sqrt{\frac{2GM}{3R}}$ (C) $V_{\min} = \sqrt{\frac{5GM}{7R}}$ (D) $V_{\min} = \sqrt{\frac{2GM}{R}}$
- Q.6 Two rods A and B of the same material and length have radii r_1 and r_2 , respectively. When they are rigidly fixed at one end and twisted by the same couple applied at the other end, the ratio $\left(\frac{\text{angle of twist at end of A}}{\text{angle of twist at the end of B}} \right)$ is :
(A) $\frac{r_1^2}{r_2^2}$ (B) $\frac{r_1^3}{r_2^3}$ (C) $\frac{r_2^4}{r_1^4}$ (D) $\frac{r_1^4}{r_2^4}$

- Q.7 There is a solid sphere of radius R and uniform mass density ρ in which there is a spherical cavity of radius $(R/4)$. The centre of spherical cavity is at a distance of $\left(\frac{R}{2}\right)$ from the centre of the solid sphere. Then the gravitational field intensity at the centre of the spherical cavity is

(A) $\frac{\pi GR\rho}{6}$ (B) $\frac{2}{3}\rho R\pi G$ (C) $\frac{\pi GR\rho}{3}$ (D) $\frac{\pi GR\rho}{4}$

- Q.8 The gravitational force of attraction between a uniform sphere of mass M and a uniform rod of length ℓ and mass m oriented as shown is



(A) $\frac{GMm}{r(r+\ell)}$ (B) $\frac{GM}{r^2}$ (C) $\frac{GMm}{r^2}$ (D) $\frac{GMm}{r}$

- Q.9 One cm^3 of water is taken from the surface to the bottom of a lake 200 m deep. If the bulk modulus of water is 2.2×10^4 atmosphere, the density 10^3 kg/m^3 and atmospheric pressure is 10^5 N/m^2 , then the change in volume is [Take $g = 9.8 \text{ m/s}^2$]
 (A) $8.9 \times 10^{-8} \text{ cm}^3$ (B) $8.9 \times 10^{-7} \text{ cm}^3$ (C) $8.9 \times 10^{-4} \text{ cm}^3$ (D) $8.9 \times 10^{-2} \text{ cm}^3$

- Q.10 In a double star system one of mass m_1 and other of mass m_2 with a separation d , rotate about their common centre of mass. Then the ratio of the areal velocity of star of mass m_1 to that of star of mass m_2 about their common C.M is

(A) $\frac{m_1}{m_2}$ (B) $\frac{m_2}{m_1}$ (C) $\frac{m_1^2}{m_2^2}$ (D) $\frac{m_2^2}{m_1^2}$

- Q.11 A geostationary satellite is revolving at a height $6R$ above the earth's surface (R being radius of earth). The period of revolution for a satellite revolving at a height $2.5 R$ above earth's surface will be (in hours)

(A) 24 (B) $12\sqrt{2}$ (C) 12 (D) $6\sqrt{2}$

- Q.12 A wire of length L and cross-section A is made of material of Young's modulus Y . It is stretched by an amount x the work done by external agent is

(A) $\frac{YxA}{2L}$ (B) $\frac{Yx^2A}{L}$ (C) $\frac{Yx^2A}{2L}$ (D) $\frac{2Yx^2A}{L}$

- Q.13 A rocket of initial mass $2m$ (rocket plus fuel) takes off in a straight line from a station in outer space. If the velocity of ejection of gas is u (relative to the rocket), the net K.E. of the rocket when its mass is m is (approximately):

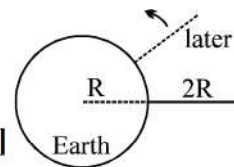
(A) $\frac{1}{2}mu^2$ (B) $\frac{1}{4}mu^2$ (C) mu^2 (D) $\frac{1}{3}mu^2$

- Q.14 Find the stress in CD.
Area of CD = 2 m^2



(A) 15 N/m^2 (B) 5 N/m^2
 (C) 20 N/m^2 (D) none of these

- Q.15 Consider a very long stick of length $2R$, which extends from just above the surface of the earth, to a radius $3R$. If initial conditions have been set up so that this stick moves in a circular orbit while always pointing radially. What is the period of this orbit. g represents acceleration due to gravity on surface of earth. [3]

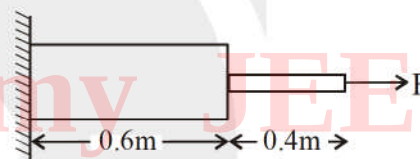


- (A) $6\pi\sqrt{\frac{R}{2g}}$ (B) $2\pi\sqrt{6}\sqrt{\frac{R}{g}}$ (C) $\pi\sqrt{6}\sqrt{\frac{R}{g}}$ (D) none of these

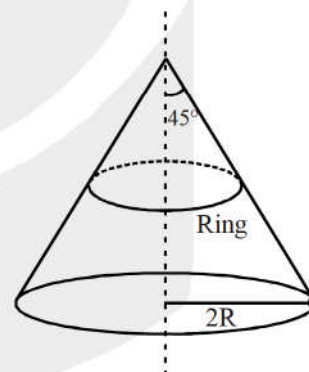
[MULTIPLE CORRECT CHOICE TYPE]

- Q.16 Earth orbiting satellite will escape if
 (A) Its speed is increased by 41 %
 (B) Its speed in the orbit is made $\sqrt{1.5}$ times of its value
 (C) Its KE is doubled
 (D) It moves in a spiraling path moving in the orbit
- Q.17 When a steel wire fixed at one end is pulled by a constant force F at its other end, its length in equilibrium increases by l . Which of the following statements is not correct?
 (A) Work done by the external force is Fl .
 (B) Some heat is produced in the wire in the process.
 (C) The elastic potential energy of the wire is $Fl/2$.
 (D) The heat produced is equal to half of the elastic potential energy of the wire.

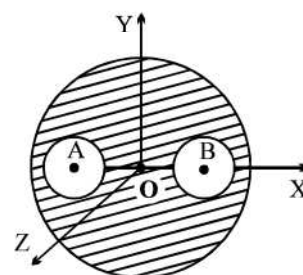
- Q.18 Two bars of steel ($Y = 2 \times 10^{11} \text{ N/m}^2$) are joined together as shown. The area of cross section of the left bar is 15 cm^2 and the area of right bar is unknown. The extension in both bars is the same.



- (A) The area of right bar is 10 cm^2
 (B) The stresses in left and right bar are in ratio 3 : 2
 (C) The decrease in thickness of bar is more for the left
 (D) The decrease in thickness of bar is more for right bar
- Q.19 A ring of mass m , radius R , cross-sectional area a and Young's modulus Y is kept on a smooth cone of radius $2R$ and semi-vertical angle 45° , as shown in the figure. Assume that the extension in the ring is small.



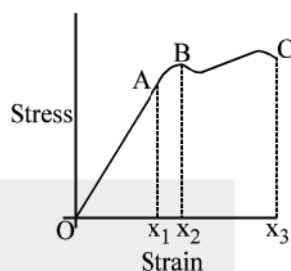
- (A) The tension in the ring will be same throughout.
 (B) The tension in the ring will be independent of the radius of ring if radius of the ring is less than $2R$.
 (C) The extension in the ring will be $\frac{mgR}{aY}$
 (D) Elastic potential energy stored in the ring will be $\frac{m^2g^2R}{8\pi Ya}$
- Q.20 A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit, with their centres at $A(-2, 0, 0)$ and $B(2, 0, 0)$ respectively are taken out of the solid leaving behind spherical cavities as shown in figure.



- (A) the gravitational force due to this object at the origin is zero
 (B) the gravitational force at the point $B(2, 0, 0)$ is zero
 (C) the gravitational potential is the same at all points of the circle $y^2 + z^2 = 36$
 (D) the gravitational potential is the same at all points of the circle $y^2 + z^2 = 4$

[MATCH THE COLUMN TYPE]

- Q.21 A metal wire is pulled by equal forces at the ends. The stress-strain (x) relationship of the wire is given. Point B of graph is the elastic limit.



Column I

- (A) $x < x_1$
 (B) $x_1 < x < x_2$
 (C) $x_2 < x < x_3$
 (D) $x > x_3$

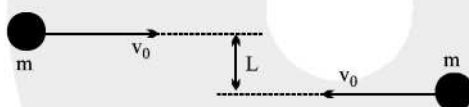
Column II

- (P) There is permanent deformation
 (Q) Hook's law is valid
 (R) There is no permanent deformation
 (S) The wire breaks

[SUBJECTIVE TYPE]

- Q.22 A metal disc of radius R , density ρ and thickness t is rotating with angular velocity ω in gravity free space. Find the tensile stress in the disc at any distance r from its centre. Hence find the total elastic strain energy in the entire disc. The young's modulus of the disc's material is Y .

- Q.23 In the figure shown, the two tiny but heavy bodies having masses ' m ' are initially very far away from each other. At the initial moment, they are imparted velocities $v_0 = \sqrt{\frac{Gm}{L}}$ as shown. Find their closest approach distance. The only force acting on them is their mutual gravitation attraction.



- Q.24 A rocket starts vertically upwards with speed v_0 from the earth surface. Find its speed v at a height h , where R is the radius of the earth. Hence deduce the maximum height reached by a rocket fired with speed equal to 90% of escape velocity.
- Q.25 A cylindrical wire of radius 1 mm and length 1 m is subjected to a tension of 100 N. Find the change in radius of the wire. Poisson's ratio for the material = 0.25, Young's modulus = $2 \times 10^{10} \text{ N/m}^2$.

ANSWER KEY

- Q.1 A Q.2 A Q.3 C Q.4 B Q.5 B Q.6 C Q.7 B
 Q.8 A Q.9 C Q.10 D Q.11 D Q.12 C Q.13 B Q.14 A
 Q.15 B Q.16 A, C Q.17 D Q.18 A, D Q.19 A, B, C Q.20 A, C, D

- Q.21 (A) Q, R; (B) R; (C) P; (D) P, S Q.22 $E = \frac{t \pi \rho^2 \omega^4 r^6}{6Y}$

- Q.23 $r = \left(\frac{\sqrt{5}-1}{2} \right) L$ Q.24 $v_0^2 - v^2 = \frac{(2gh)}{\left(1 + \frac{h}{r} \right)}, \frac{81}{19} R$ Q.25 $\Delta r = \frac{1.25}{\pi} \times 10^{-6} \text{ m}$